INCH POUND

MIL-B-42079
31 May 1994

MILITARY SPECIFICATION

BRACE ASSEMBLY

Adjustable Bed

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a stabilizing brace that attaches to the Adjustable Bed without the use of tools to increase rigidity. Two stabilizing braces (one pair) are used for each bed. (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-MSE, 2800 South 20th Street, Philadelphia, PA 19145, by using the self-addressed Standardization Document Improvement Proposal, (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 6530

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SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
A-A-883	-	Tape, Pressure Sensitive Adhesive, Masking
P-C-436	-	Cleaning Compound, Alkali, Boiling Vat (Soak) or Hydrosteam
QQ-A-200/8	-	Aluminum Alloy 6061, Bar, Rod, Shapes, Tube, and Wire, Extruded
QQ-A-225/8	-	Aluminum Alloy 6061, Bar, Rod, Wire, and Special Shapes, Rolled, Drawn, or Cold Finished
QQ-A-591	-	Aluminum Alloy Die Castings
QQ-S-763	-	Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting
WW-T-700/6	-	Tube, Aluminum Alloy, Drawn, Seamless, 6061
PPP-B-566	-	Box, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated, Plywood
PPP-B-621	-	Box, Wood, Nailed and Lock-Corner
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-10547	-	Liner Case and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible
MIL-B-36558	-	Bed, Adjustable

STANDARDS

FEDERAL

FED-STD-151	-	Metal: Test Methods
FED-STD-595	-	Color

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-129-2	-	Marking for Shipment and Storage - Medical Material
MIL-STD-147	-	Palletized Unit Loads
MS-16562	-	Pin, Spring, Tubular, Slotted

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from:

Defense Printing Service Detachment Office
700 Robbins Avenue
Building 4D
Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

DRAWING No. 20418FD, Brace Assembly

(Application for copies of the drawing should be addressed to Defense Personnel Support Center, ATTN: DPSC-MST, 2800 So. 20th Street, Philadelphia, PA 19145.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AWS A5.10-80 - Specification for Aluminum and Aluminum Alloy Bare Welding Rods and Electrodes

(Application for copies should be addressed to:

American National Standards Institute
11 West 42nd Street
New York, NY 10036)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging

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(Application for copies should be addressed to:

American Society for Testing and Materials
1916 Race Street
Philadelphia, PA 19103-1187)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Material. Materials shall conform to applicable portions of specifications, standards, and drawings listed in section 2 and shall be subjected to tests and meet requirements specified therein. Metals shall conform to chemical requirements of their applicable specifications when tested as specified in 4.4.4.

3.2.1 Tubes. Tubes shall be made of aluminum-alloy tubing, 6061-T6, 0.500 inch outside diameter, 0.065 inch thick wall conforming to QQ-A-200/8, QQ-A-225/8, or WW-T-700/6.

3.2.2 Yoke, blade, knuckle, male clamp, female clamp, and cup assembly. Yoke, blade, knuckle, male clamp, and female clamp shall be machined from 6061-T6 aluminum alloy conforming to QQ-A-200/8 or QQ-A-225/8, or shall be die cast in conformance QQ-A-591 from aluminum alloy number 360.0. If cup assembly is of welded construction, the welding electrodes shall be type ER4043, conforming to ANSI/AWS A5.10-80.

3.2.3 Clamp screw. Clamp screw shall be corrosion resisting steel conforming to QQ-S-763, type 304.

3.2.4 Drive screw. Drive screws shall be corrosion-resistant steel, type 18-8 and be 3/16 inch long; .086 inch diameter with round head.

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3.3 Construction. Construction shall conform to drawing 20418FD. Each brace assembly shall be assembled from three yokes, one blade, one knuckle, two clamp assemblies, one cup assembly, and one long and one short tube. Brace assembly shall be tested as specified in 4.6.1. Installation procedure for the brace assembly shall be as shown in figure 1.

3.4 Finish.

3.4.1 Powder coating. An epoxy powder formulated for electrostatic spray equipment application shall be used as the finish on metal surfaces. The epoxy coating shall be fast curing and shall provide superior adhesion with a tough, mar-resistant, smooth finish that withstands high impact without chipping or cracking. The coating shall be olive drab in color conforming to No. 24087 of Fed. Std. 595. See 6.5 for availability.

3.5 Identification marking. The cup of each brace assembly shall be permanently and legibly marked with name or registered trademark of manufacturer.

3.6 Workmanship. Brace assembly shall be free from specified or unspecified defects that detract from appearance or impair serviceability.

3.7 Fit to adjustable bed. When installed as described in 4.6.1, the braces shall securely attached to the bed. The foot of the bed shall fully seat in the bottom of the cup of the brace assembly. The clamps shall fit snugly around the leg and side rail of the bed when the clamp screws are tightened by hand.

3.8 Load bearing capability. When tested in accordance with 4.6.2, the braces shall show no signs of permanent deformation of any parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable or the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory and investigative without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2.1 Component and material inspection. The contractor is responsible for ensuring components and materials are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.2.2 Disassembly inspection. Failure of first article to pass any examination or test shall be cause for disassembly in the presence of a Government representative. Each disassembled part shall be examined in detail for compliance with this specification and Drawing 20418FD. Nonconforming parts shall be rejected and shall be cause for rejection of first article. Reassembly with replacement parts and retesting shall be responsibility of contractor.

4.2.3 Certificates of quality. Certificates of quality, supplied by manufacturer of components and materials used in the brace assembly, may be furnished in lieu of actual testing by contractor. Certificates shall list test methods and results.

4.2.4 Classification of inspection. Inspections are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.7).

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4.3 First article inspection.

4.3.1 Examination. First article brace assembly shall be examined as specified in 4.5. Presence of one or more defects in the brace assembly shall be cause for rejection.

4.3.2 Tests. First article brace assembly shall be tested as specified in 4.6.1 and 4.6.2. Failure of test shall be cause for rejection and performing disassembly inspection specified in 4.2.2.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 at the inspection levels listed in table 1.

TABLE I. Sampling.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects (each component).	II	1.0
Minor defects (each component).	II	2.5
For dimensional examination.	S-3	2.5

4.4.2 Sampling for tests. Samples selected in accordance with 4.4.1 shall be examined and tested as specified in 4.5 and 4.6. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Test requirements.

Item	Inspection level	Percent defective AQL	Minimum sample size
Component material testing	S-1	*	3
End item testing	S-2	1.5	

* Acceptance number shall be zero for all sample sizes.

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4.4.3 Components. Samples of components shall be selected from each lot of components used in manufacturing brace assemblies. The same component may be used to perform two or more inspections or tests provided it is not destroyed or damaged by a previous test.

4.4.4 Chemical testing of metals. Four ounces of each lot of each metal used in construction of brace assemblies shall be selected for chemical analysis in accordance with methods 111 or 112 of FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

4.5 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. A dimension that is not within the tolerance specified on the drawing is classified as a defect.

TABLE III. Categories and defects.

Major A Category	Brace assembly (PL20418) Defect
101	Brace assembly not complete with all component parts
102	Components not properly assembled or aligned
103	Exposed and accessible edges and surfaces not free from sharp edges and burrs
104	Center-to-center distance between cup assembly and clamp assemblies not within tolerances specified.
Major B Category	**Clamp assembly Defect**
101	Clamp halves do not hinge freely
102	Tubular spring pin loose or not flush
103	Clamp screw binds

4.6 Performance tests on brace assemblies. Inability to perform one or more of the following performance tests as stated shall constitute failure of performance testing. An adjustable bed meeting the requirements of MIL-B-36558 shall be used for these tests.

4.6.1 Fit test to bed. Two brace assemblies shall be installed to the bed in accordance with the instructions of figure 1 of this specification. Braces shall be installed at diagonally opposite corners of the bed.

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4.6.2 Load test. The bed with braces installed at diagonally opposite corners shall be loaded with 900 pounds evenly distributed over the load bearing area and allowed to remain for 30 minutes.

4.7 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Unit. One package containing one pair of aluminum brace assemblies and one aluminum cup receptacle, together with instructions and/or a diagram for installation, as specified, constitutes one unit.

5.1.1.1 Cleaning and drying. Each unit (one pair of brace assemblies and one cup receptacle) shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.2 Level A.

5.1.2.1 Unit package. Each pair of brace assemblies shall be wrapped in a single 36 inch by 60 inch sheet of untreated paper folded in such a way to separate and protect each brace assembly individually. As an alternate, each brace assembly shall be wrapped individually in an 18 inch by 60 inch sheet of untreated paper. The wrapped brace assemblies shall be nested as shown in Figure 2 and shall be packaged in a box conforming to PPP-B-636, type CF, class domestic, PPP-B-566, or PPP-B-676. Inside dimensions of the box shall be 2 inches by 6 inches by 42 inches. Box design shall include interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents.

5.1.3 Level C. Preservation for each unit (one pair of brace assemblies and one cup receptacle) shall conform to the requirements of ASTM D 3951.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

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5.2.1 Level A. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather resistant. Closure, strapping and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. Four unit packages shall be packed in accordance with the requirements of ASTM D 3951.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceeds 250 lbs (excluding the pallet) or 20 cubic feet. Loads shall be unitized as specified in MIL-STD-147, on type IV or type V 4-way entry pallets. Pallet shall have a length of 40 inches and width of 48 inches. Pallet load, including the pallet, shall not exceed 54 inches in height, 43 inches in length and 52 inches in width. Quantity for shipment to one destination of less than 250 lbs or 20 cubic feet need not be palletized.

5.2.5.1 Levels A and B. All units packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. All units packed as specified in 5.2.3 shall be unitized as specified in ASTM D 3951.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit, exterior (shipping) container and unitized load shall be marked as specified in MIL-STD-129-2.

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5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.7.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The brace assembly is intended to be used as an accessory to the Adjustable Bed, NSN: 6530-01-015-5189. Two brace assemblies (one pair) installed on diagonally opposite corners of the bed provide maximum stabilization.

6.2 Acquisition requirements.

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. First article submission date (see 3.1).

d. Number of brace assemblies to be submitted for first article inspection (see 6.3).

e. When and where the Government will conduct tests and examinations (see 4.1).

f. Packaging and packing levels (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, four brace assemblies shall be a first article sample. Contracting officer shall include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article tests results and disposition of first articles. Invitations for bids shall provide that the Government reserves the right to waive requirement samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract.

6.4 Availability of color chips. The individual 3 by 5 inch chips of the color referenced in 3.6.2 is available from Commanding Officer, U.S. Naval depot, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 Powder coating. A suitable epoxy coating powder is Corvel ECA-1555-GP2, available from the Polymer Corporation, Reading, PA, or equal.

6.6 Subject term (key word) listing.

1. Reinforcers
2. Stabilizers
3. Support structure

6.7 National stock number coverage. The brace assembly covered in this specification is listed as follows in the Federal Supply System:

National Stock Number	Item Identification
6530-01-328-5194	Brace Assembly, (Adjustable Bed)

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Custodians:

Review activities:

Air Force - 03
Army - MD
Navy - MS

Preparing activity:

DoD - MB

Agent:

DLA - DM

Project No. 6530-2387
Location: BRACE13M.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-42079	2. DOCUMENT DATE (YYMMDD) 940531

3. DOCUMENT TITLE
BRACE ASSEMBLY, Adjustable Bed

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defénse Personnel Support Center ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial 215-737-2870	(2) AUTOVON 444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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